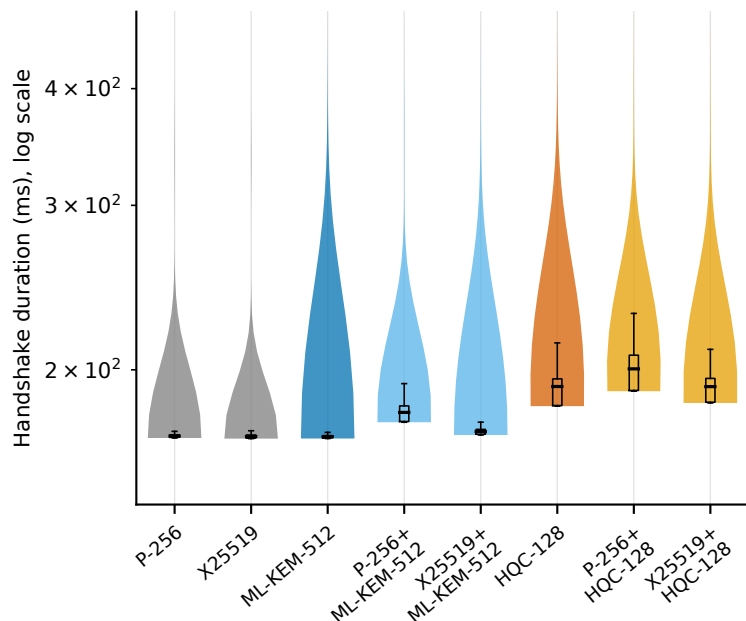
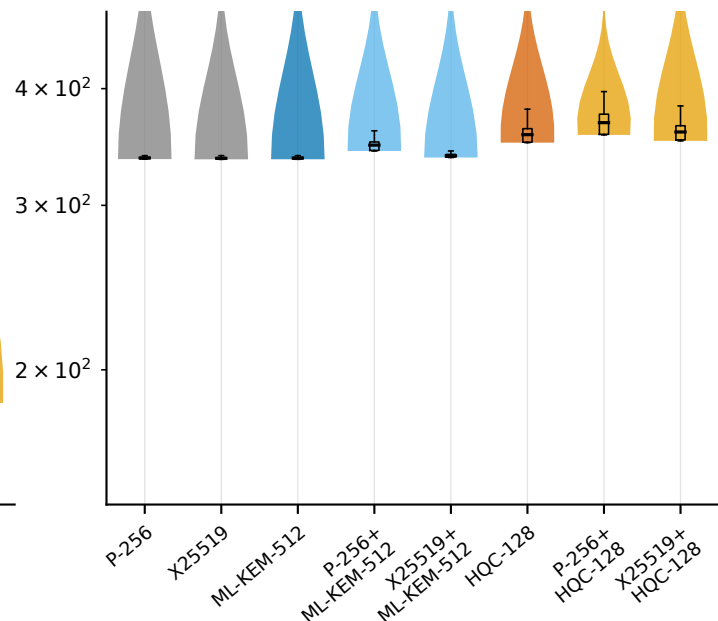


**(a)**

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**

Classical KEX
  ML-KEM (pure)
  ML-KEM (hybrid)
  HQC (pure)
  HQC (hybrid)

| KEM               | ed25519<br>Mean (ms) | ed25519<br>CV (%) | ed25519<br>Max (ms) | mldsa44<br>Mean (ms) | mldsa44<br>CV (%) | mldsa44<br>Max (ms) |
|-------------------|----------------------|-------------------|---------------------|----------------------|-------------------|---------------------|
| P-256             | 181.3                | 55.4              | 1.17k               | 390.4                | 75.0              | 2.44k               |
| X25519            | 177.1                | 44.4              | 1.17k               | 374.6                | 64.7              | 2.44k               |
| ML-KEM-512        | 208.4                | 108.4             | 2.14k               | 378.4                | 67.9              | 2.37k               |
| P-256+ML-KEM-512  | 195.9                | 67.0              | 1.38k               | 378.9                | 59.7              | 2.38k               |
| X25519+ML-KEM-512 | 206.6                | 95.5              | 1.37k               | 371.1                | 61.2              | 2.37k               |
| HQC-128           | 224.8                | 95.5              | 1.65k               | 395.2                | 52.9              | 1.86k               |
| P-256+HQC-128     | 221.3                | 77.7              | 1.66k               | 390.5                | 37.7              | 1.87k               |
| X25519+HQC-128    | 213.2                | 80.8              | 1.65k               | 401.9                | 56.9              | 2.42k               |